

Tejas Nasa

+91 9599875310 | tejasnasa1908@gmail.com | linkedin.com/in/tejasnasa | github.com/tejasnasa | [portfolio](#)

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, C++, SQL, HTML5, CSS3
- **Backend:** Node.js, Express.js, FastAPI, Socket.io, Yjs (CRDTs), Celery, Supabase, Appwrite, BetterAuth, Zod
- **Frontend:** React.js, Next.js, Astro, TanStack Query, TanStack Router, Redux, Zustand, Tailwind, Recharts
- **DevOps & Tooling:** Git, GitHub, Postman, Docker, GitHub Actions, Nginx, Digital Ocean, Azure, Vercel
- **Database:** PostgreSQL, MongoDB, Redis, pgvector, Prisma, Drizzle, SQLAlchemy, Alembic

EXPERIENCE

Psync - SWE Intern

Jun. 2026 – Present

Astro, FastAPI, React.js, Appwrite, Tailwind CSS, Tree-Sitter, Redis, Pydantic

Remote

- Migrated the authentication system from JWT-based auth to Appwrite's cookie-based session model, rebuilding all 7 auth flows: login, signup, logout and Google/GitHub OAuth across the application.
- Created a repo-aware RAG pipeline for autonomous issue resolution: Tree-sitter chunking at function/class granularity, OpenAI embeddings stored in per-user Redis, and KNN-grounded LLM plans that reference real files/functions, with incremental re-indexing via a 6h drift ticker that re-embeds only changed files.
- Rebuilt the platform's entire UI from scratch in Astro using React, Tailwind CSS, and DaisyUI, redesigning core dashboards and building 10+ chart types with Chart.js, including custom visualizations to deliver a more modern, consistent, and intuitive interface.

PROJECTS

Illume - AI-Powered Codebase Onboarding Platform

[GitHub](#) | [Live Link](#)

FastAPI, Next.js, Postgres, SQLAlchemy, pgvector, Tree-sitter, Celery, React Force Graph

- Engineered a topological learning engine using FastAPI and Tree-sitter, executing Kahn's algorithm across parsed AST dependency graphs to generate customized codebase reading paths for 500+ files.
- Architected a multi-source RAG pipeline utilizing Celery, vectorizing 5,000+ Git commits, pull requests, and semantic code nodes into pgvector to answer deep architectural queries with traceable source citations.
- Built an interactive Next.js 15 onboarding dashboard, visualizing 1,000+ node architectures in 3D while instantly identifying critical file bus-factors and streaming real-time ingestion telemetry via WebSockets.
- Deployed the backend to an Azure VM via a Docker + GitHub Actions CI/CD pipeline, with Nginx as a reverse proxy, handling SSL termination and WebSocket upgrades.

Nimbus - Real Time Collaborative Workspace

[GitHub](#) | [Live Link](#)

Node.js, Socket.io, Redis, Postgres, Yjs, Next.js, Excalidraw, Express, Prisma, WebRTC

- Architected a horizontally scalable WebSocket microservice using Node.js and Socket.io, integrating a Redis Pub/Sub adapter to concurrently handle 1,000+ state broadcasts/min with low-latency delivery.
- Implemented a CRDT engine with Yjs, resolving multi-client merges via incremental binary state updates and eliminating 90% of state-syncing conflicts across rich-text (Markdown) and canvas (Excalidraw) surfaces.
- Built an AI agent that generates Markdown documents and Excalidraw whiteboards from prompts using Qwen and GPT-OSS, streaming content live into the shared canvas for all collaborators.
- Engineered a monorepo-based workspace in Next.js supporting 5+ concurrent editors, with native WebRTC voice rooms (self-hosted Coturn relay) and Better-Auth access control within a unified Turborepo pipeline.

EDUCATION

Bachelor of Technology in Information Technology

Maharaja Surajmal Institute of Technology, Delhi

2023 – 2027

CGPA: 8.4/10